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IGCSE Mathematics · exam-style revision

Worked Solutions

MATHEMATICS **NNA 0580/04**

Paper 4 Calculator (Extended) **October/November 2026**

Full Revision Paper 1 · Worked Solutions **100 marks**

HOW THE MARKS ARE SHOWN

- **M1** method mark — awarded for a correct method, even if the arithmetic that follows is wrong.
- **A1** accuracy mark — the correct value. It depends on the method mark being earned.
- **B1** independent mark — awarded for the answer alone, with no method required.
- **FT** follow through — a later part is marked on your own earlier (wrong) value, so never stop after one mistake.

ACCURACY

- Give answers to **3 significant figures**, and angles to **1 decimal place**, unless the question says otherwise.
- Never round part way through. Keep the full value in the calculator and round once, at the end.
- In a **show that** question you must display more accuracy than the printed answer before you round to it.

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[Turn over

Worked solutions

1 **Solve $4(3x - 5) = 2(x + 9) + 2$** [3]

Expand both brackets: $12x - 20 = 2x + 18 + 2$ **M1**

$12x - 20 = 2x + 20$

Collect terms: $10x = 40$ **M1**

$x = 4$

> *Expand first, then collect. Dividing by 4 before expanding only works if every term on both sides is divisible by 4 — here the right-hand side is not.*

2 **Work out $\dfrac{16.71 + 46.13}{\sqrt{8.6^2 - 3.5^2}}$ to 2 significant figures** [2]

Top: $16.71 + 46.13 = 62.84$

Bottom: $8.6^2 - 3.5^2 = 73.96 - 12.25 = 61.71$, so $\sqrt{61.71} = 7.85557\ldots$ **M1**

$62.84 \div 7.85557\ldots = 7.99942\ldots$

8.0

> *Put the whole numerator and the whole denominator in brackets on the calculator. The answer is **8.0**, not 8 — 2 significant figures needs the zero.*

3 **A printer is reduced by 18% to \$287. Find the original price.** [2]

\$287 is not the original price; it is 82% of it.

original × 0.82 = 287, so original = 287 ÷ 0.82 **M1**

\$350

> *Going forward you multiply; coming back you divide. Check: 350 × 0.82 = 287 ✓. Finding 18% of 287 and adding it on gives \$338.66, which fails this check.*

4 **Share \$4560 in the ratio 5 : 7 : 12. Find Chen's amount.** [2]

Total parts: $5 + 7 + 12 = 24$

One part = $4560 \div 24 = 190$ **M1**

Chen = 12×190

\$2280

> *Always find one part first. It also gives a check: 950 + 1330 + 2280 = 4560 ✓.*

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5 **Sequence 2, 6, 12, 20, 30, …** [3]

First differences: 4, 6, 8, 10 — not constant.

Second differences: 2, 2, 2 — constant, so the sequence is quadratic.

(a) Next term = 30 + 12 = 42 **B1**

(b) Second difference 2, so the n^2 coefficient is 2 ÷ 2 = 1. **M1**

Subtract n^2 from each term: 1, 2, 3, 4, 5 — that is n.

$n^2 + n$ (= $n(n + 1)$)

Test it: n = 1 gives 2 ✓ and n = 4 gives 20 ✓. Always check two terms before moving on.

6 **Factorise fully $6x^2 - 7x - 3$** [2]

6 × −3 = −18. Two numbers multiplying to −18 and adding to −7: −9 and +2. **M1**

$6x^2 - 9x + 2x - 3 = 3x(2x - 3) + 1(2x - 3)$

$(3x + 1)(2x - 3)$

Expand your answer to check — it takes ten seconds and catches every sign error.

7 **$950 at 3.4% per year compound interest for 6 years.** [3]

Compound interest is one multiplier raised to a power.

950×1.034^6 **M2**

$= 950 \times 1.222146\ldots = 1161.039\ldots$

$1161

*Do **not** work out six separate years on rounded figures — you will drift by a dollar or two and lose the accuracy mark. One multiplier, one power, one rounding at the end.*

8 **$1330 at r% per year simple interest reaches $1968.40 after 8 years.** [3]

Interest earned = 1968.40 − 1330 = $638.40 **M1**

Simple interest is the same every year: 638.40 ÷ 8 = $79.80 **M1**

As a percentage of the starting $1330: $\frac{79.80}{1330} \times 100$

$r = 6$

Subtract the principal first. Dividing 1968.40 by 1330 gives the total growth factor 1.48, not the yearly rate.

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9 **Solve $3x + 2y = 4$ and $5x - 4y = 25$** [3]

The y terms are +2 and −4, so double the first equation to match them.

$\times 2$: $6x + 4y = 8$ **M1**

Signs are opposite, so **add**: $11x = 33$, $x = 3$ **A1**

Substitute: $9 + 2y = 4 \rightarrow 2y = -5$

$x = 3,\ y = -2.5$

Check in the other equation: $5(3) - 4(-2.5) = 15 + 10 = 25$ ✓.

10 **Simplify fully** $\dfrac{3x^2 - 12}{x^2 - x - 6}$ [3]

Numerator: take out 3 first $\rightarrow 3(x^2 - 4) = 3(x - 2)(x + 2)$ **B1**

Denominator: two numbers × to −6, + to −1 $\rightarrow$ −3 and +2

$= (x - 3)(x + 2)$ **B1**

Cancel the common factor $(x + 2)$.

$$\frac{3(x-2)}{x-3}$$

Take the number out first — $3x^2 - 12$ does not look like a difference of two squares until you do. Never cancel across a + or − sign.

11 **Make x the subject of** $y = \dfrac{3x + 2}{x - 5}$ [4]

Multiply up: $y(x - 5) = 3x + 2$ **M1**

Expand: $xy - 5y = 3x + 2$ **M1**

Collect every x term on one side: $xy - 3x = 5y + 2$ **M1**

Factorise: $x(y - 3) = 5y + 2$

$$x = \frac{5y + 2}{y - 3}$$

*When the new subject appears twice you **must** gather both terms and take it out as a common factor. Nothing else works.*

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12 The interior angle of a regular polygon is 162°. Find the number of sides. [2]

Go via the exterior angle — always the quicker route.

Exterior angle = 180 − 162 = 18° **M1**

Exterior angles of any polygon total 360°: $\frac{360}{18}$

20 sides

13 $y = x^3 - 6x^2 + 9x + 2$ — find the two turning points and their nature. [5]

Differentiate: $\frac{dy}{dx} = 3x^2 - 12x + 9$ **M1**

Turning points have zero gradient, so set it to 0:

$3(x^2 - 4x + 3) = 0 \rightarrow 3(x - 1)(x - 3) = 0$ **M1**

$x = 1$ or $x = 3$ **A1**

Substitute into the **original** equation:

$x = 1$: $1 - 6 + 9 + 2 = 6$ $x = 3$: $27 - 54 + 27 + 2 = 2$ **A1**

(1, 6) maximum · (3, 2) minimum

> *A positive cubic rises, turns down, then rises again, so the **first** turning point is the maximum. Or test the gradient either side: at x = 0 it is +9, at x = 2 it is −3 — going + → 0 → − is a maximum. Substituting back into the derivative instead of the curve gives 0 and 0, which should warn you.*

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14 Speed–time graph: acceleration, distance and average speed. [5]

(a) Acceleration is the **gradient**: $\frac{18 - 0}{12} = 1.5$ **B1**

(b) Distance is the **area under** the graph — a trapezium. **M1**

Parallel sides: top = 42 − 12 = 30 s, bottom = 60 s. Height = 18 m/s.

½(30 + 60) × 18 = ½ × 90 × 18 = 810 **M1**

(c) Average speed = $\frac{810}{60} = 13.5$ **B1**

(a) 1.5 m/s^2 · (b) 810 m · (c) 13.5 m/s

*The top side runs from 12 s to 42 s, so it is **30**, not 42 — the most common lost mark on travel graphs. Or split into triangle 108 + rectangle 540 + triangle 162 = 810 ✓. Units tell you which is which: m/s per s is acceleration, m/s × s is distance.*

15 $24 000 losing 12% of its value each year. [4]

(a) Losing 12% leaves 88%, so the multiplier is 0.88. **M1**

$24\,000 \times 0.88^5 = 24\,000 \times 0.527731\ldots = 12\,665.566\ldots$ **A1**

(b) Try successive powers until the value drops below 8000: **M1**

8 years: $24\,000 \times 0.88^8 = \$8631$ 9 years: $24\,000 \times 0.88^9 = \$7595$

(a) $12 665.57 (≈ $12 700 to 3 s.f.) · (b) 9 years

*Show the year **before** as well — the 9-year value alone does not prove that 8 years was still above $8000. With logs: $\frac{\log(8000 \div 24000)}{\log 0.88} = 8.59$, so 9 whole years.*

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16 OA = $\boldsymbol{a}$, OB = $\boldsymbol{b}$, M the midpoint of AB, N on OB with ON : NB = 3 : 1. [5]

(a) A to B: back down $\boldsymbol{a}$, then out along $\boldsymbol{b}$.

$\mathbf{AB} = -\boldsymbol{a} + \boldsymbol{b} = \boldsymbol{b} - \boldsymbol{a}$ **B1**

(b) $\mathbf{OM} = \boldsymbol{a} + \frac{1}{2}(\boldsymbol{b} - \boldsymbol{a})$ **M1**

$= \boldsymbol{a} - \frac{1}{2}\boldsymbol{a} + \frac{1}{2}\boldsymbol{b} = \frac{1}{2}(\boldsymbol{a} + \boldsymbol{b})$ **A1**

(c) ON : NB = 3 : 1 means N is $\frac{3}{4}$ of the way along OB, so $\mathbf{ON} = \frac{3}{4}\boldsymbol{b}$ **M1**

$\mathbf{MN} = -\mathbf{OM} + \mathbf{ON} = -\frac{1}{2}\boldsymbol{a} - \frac{1}{2}\boldsymbol{b} + \frac{3}{4}\boldsymbol{b}$

$\mathbf{AB = b - a}$ · $\mathbf{OM = \frac{1}{2}(a + b)}$ · $\mathbf{MN = \frac{1}{4}b - \frac{1}{2}a}$

> *One rule solves all of these: to get from X to Y, go back to O and forward again — **XY** = −**OX** + **OY**. A 3 : 1 ratio means **quarters**, not thirds: there are 3 + 1 = 4 parts.*

17 Transformations on the grid. [5]

(a) A is (1, 1), (3, 1), (1, 4); B is (1, −1), (1, −3), (4, −1).

Same size and shape but turned, so it is a **rotation**. **B1**

The rule $(x, y) \to (y, -x)$ is a quarter turn clockwise. **B1**

The origin is the only point that does not move. **B1**

(b) From the centre (0, 1), go to each vertex then twice as far the **other** way: **M1**

(1, 1) → (−2, 1) (3, 1) → (−6, 1) (1, 4) → (−2, −5) **A1**

(a) Rotation, 90° clockwise, centre (0, 0)

> *"Describe fully" needs all three facts: the word, the angle with its direction, and the centre. Naming two transformations scores **zero** — the question says single. A negative scale factor puts the image upside down on the opposite side of the centre.*

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18 Circle theorems: AC is a diameter, angle BAC = 34°, angle CAD = 28°. [5]

(a) B is on the circle and AC is a diameter, so ABC is the angle in a semicircle. **B1**

Angle ABC = 90° *Reason: the angle in a semicircle is a right angle.* **B1**

(b) Angles in triangle ABC add to 180°: 180 − 90 − 34 **B1**

Angle BCA = 56°

(c) The whole angle at A: angle BAD = 34° + 28° = 62° **M1**

ABCD is a cyclic quadrilateral, so A and C are opposite: 180 − 62

ABC = 90° · BCA = 56° · BCD = 118° (opposite angles of a cyclic quadrilateral)

> ***No reason, no mark.*** *Name the theorem in full. "Angle in a semicircle" scores; "because AC is a diameter" does not. Part (c) needs angle BA**D**, not BAC — add the two given angles first.*

19 $y = x^3 - 3x - 1$ — table, graph and tangent. [9]

(a) $x = -1.5$: $(-1.5)^3 - 3(-1.5) - 1 = -3.375 + 4.5 - 1 =$ **0.125** **B1**

$x = 1.5$: $3.375 - 4.5 - 1 =$ **−2.125** **B1**

(b) Plot all ten points and join with one smooth curve. The curve turns near (−1, 1) and (1, −3). **B4**

(c) Draw a tangent touching the curve at (2, 1) — one straight line resting on the curve, not cutting it. **M1**

Take two points far apart on **your** tangent: gradient = $\frac{\text{rise}}{\text{run}}$ **M1**

(a) 0.125 and −2.125 · (c) gradient ≈ 9

> *$(-1.5)^3 = -3.375$, not +3.375 — use brackets on the calculator. For the gradient anything from about 7.5 to 10.5 scores, because it depends on your own tangent; the exact value is $3(2)^2 - 3 = 9$. Quoting 9 from calculus with **no tangent drawn** scores zero for the method.*

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20 AB = 84 m, BC = 67 m, angle ABC = 108°. [10]

(a) Two sides and the angle **between** them — cosine rule. **M1**

$AC^2 = 84^2 + 67^2 - 2(84)(67)\cos 108°$ **M1**

$= 7056 + 4489 - 11256 \times (-0.309017\ldots)$

$= 7056 + 4489 + 3478.3\ldots = 15\,023.3\ldots$ **M1**

(b) Now a side and its opposite angle — sine rule. **M1**

$$\frac{\sin \text{BAC}}{67} = \frac{\sin 108°}{122.5696\ldots}$$ **M1**

$$\sin \text{BAC} = \frac{67 \times 0.951056\ldots}{122.5696\ldots} = 0.51987\ldots$$

(c) A bearing is measured clockwise from north: 052 + 31.3 **B1**

(d) The mast is vertical, so this is a right-angled triangle:

$$\tan\theta = \frac{30}{122.5696\ldots} = 0.244764\ldots$$ **M1**

(a) AC = 122.6 m · (b) 31.3° · (c) 083.3° · (d) 13.8°

*cos 108° is **negative**, so the −2bc cosA term adds on — if AC came out smaller than 84 you dropped the minus sign. Check the calculator says **DEG**: in radians AC = 129.9, a plausible-looking number that scores zero. Keep 122.5696… stored; parts (c) and (d) both use it. In (d) the horizontal distance is **AC**, not 84 or 67.*

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21 7 red and 5 blue counters, taken without replacement. [9]

(a) First counter: 12 in the bag — R is $\frac{7}{12}$, B is $\frac{5}{12}$. **B1**

Second counter: only 11 left, and one colour is one fewer.

After R: R = $\frac{6}{11}$, B = $\frac{5}{11}$ After B: R = $\frac{7}{11}$, B = $\frac{4}{11}$ **B1**

(b) Multiply **along** the top branch: $\frac{7}{12} \times \frac{6}{11} = \frac{42}{132}$ **M1**

(c) Two ways: RB or BR.

$\frac{7}{12} \times \frac{5}{11} + \frac{5}{12} \times \frac{7}{11} = \frac{35}{132} + \frac{35}{132} = \frac{70}{132}$ **M1**

(d) "At least one" — do the opposite. The only way to fail is all three red.

P(RRR) = $\frac{7}{12} \times \frac{6}{11} \times \frac{5}{10} = \frac{7}{44}$ **M2**

P(at least one blue) = $1 - \frac{7}{44}$ **M1**

(b) $\frac{7}{22}$ · (c) $\frac{35}{66}$ (0.530) · (d) $\frac{37}{44}$ (0.841)

*Along a branch you multiply; between branches you add. Without replacement both numbers change — the denominator 12 → 11 → **10** and the numerator of the colour taken. Listing "at least one blue" directly needs seven products; the complement needs one.*

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22 120 students, grouped times: mean, histogram, cumulative frequency, IQR. [11]

(a) Midpoints (each student sits at the centre of its class): 10, 25, 32.5, 37.5, 50 **M1**

$\Sigma fx = 8(10) + 22(25) + 30(32.5) + 34(37.5) + 26(50)$ **M1**

$= 80 + 550 + 975 + 1275 + 1300 = 4180$ **M1**

Mean $= \frac{4180}{120} = 34.833\ldots$

(a) 34.8 minutes

(b) Unequal widths, so the bar height is **frequency density** = frequency ÷ class width. **M1**

Class	Freq	Width	Height
$0 < t \leqslant 20$	8	20	**0.4**
$20 < t \leqslant 30$	22	10	**2.2**
$30 < t \leqslant 35$	30	5	**6**
$35 < t \leqslant 40$	34	5	**6.8**
$40 < t \leqslant 60$	26	20	**1.3**

Draw the last three bars at heights 6, 6.8 and 1.3. **A2**

(c) Running totals: 8, 30, 60, 94, 120 **B1**

Plot at the **upper** boundary — (20, 8), (30, 30), (35, 60), (40, 94), (60, 120) — from (0, 0). **B1**

(d) Lower quartile at the 30th value → $t \approx 30$; upper quartile at the 90th value → $t \approx 39.4$ **M1**

(b) 6, 6.8, 1.3 · (c) 8, 30, 60, 94, 120 · (d) IQR ≈ 9.4 minutes

> *On a histogram **area** is frequency, not height — plotting 30, 34 and 26 scores zero. Check: 20(0.4) + 10(2.2) + 5(6) + 5(6.8) + 20(1.3) = 120 ✓. Quartiles on a cumulative frequency curve use n, not n + 1, and points are plotted at the **end** of each class, never the middle.*

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